

Lab 5 — Why Did the System Behave That Way?

Friday, November 20

Last updated: October 8, 2026.

Date is on the [schedule](#). Same day: [Quiz 9](#) (inverses / rank / determinants — **not** eigenvalues) and a short [Lecture 37](#), then this lab.

Credit: Lab points are for effort in class, not for a Canvas file upload. Work the hub exercises in Octave or Python on a laptop. On lab day you get a **self-contained one-pager** with the exercises written out, space for answers and new syntax, and a blank for instructor initials after a short check-in. Preview the [one-pager](#); write on the hard copy handed out in class. Eigenvalue fluency itself is assessed mainly via [Quiz 10](#), not Quiz 9.

Resources: Lachniet Ch. 5 (eigenvalues/eigenvectors) · [Syntax sheets](#) · [Resources](#) · Lectures [35–37](#) · [Strang §6.1](#).

What we are doing

Compute eigenpairs with software, verify $Av \approx \lambda v$, use them to **explain** the Lab 4 mystery (growth vs decay by start direction), interpret complex eigenvalues as rotation/oscillation, and observe orthogonal eigenvectors for a symmetric / normal-mode style matrix.

Learning objectives

1. Compute eigenvalues and eigenvectors of a general matrix with software.
2. Verify $Av \approx \lambda v$ numerically.
3. Use dominant eigenbehavior to explain a simulated system.
4. Interpret complex eigenvalues as rotation/oscillation behavior.
5. Explore a symmetric-matrix or normal-mode example and observe orthogonal eigenvectors.

Math connections

Eigenvectors as persistent directions ([Lecture 35](#)); characteristic equation and complex eigenvalues ([Lecture 36](#)); software + dominant behavior + normal-mode demo ([Lecture 37](#)). Lab 4 recorded the mystery without eigenvalues; today we close that loop.

Key syntax (this lab)

Task	Octave	Python
Eigenvalues only	<code>eig(A)</code>	<code>np.linalg.eigvals(A)</code>
Eigenpairs	<code>[V,D] = eig(A)</code>	<code>vals, V = np.linalg.eig(A)</code>
k -th eigenvalue	<code>D(k,k)</code>	<code>vals[k]</code> (0-based)
k -th eigenvector	<code>V(:,k)</code>	<code>V[:,k]</code>
Check $Av - \lambda v$	<code>A*V(:,1) - D(1,1)*V(:,1)</code>	<code>A @ V[:,0] - vals[0]*V[:,0]</code>
Dot product	<code>v1'*v2</code>	<code>v1 @ v2</code> (1D)

Indexing reminder: Boyd x_1 is 1-based. Octave is 1-based; NumPy is 0-based. Eigenvector scaling is arbitrary (software may normalize).

Demo (optional — run once)

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Octave:

```
A = [2 1; 1 2];  
[V,D] = eig(A)  
A*V(:,1) - D(1,1)*V(:,1)    % near zero
```

Python:

```
import numpy as np  
A = np.array([[2., 1.], [1., 2.]])  
vals, V = np.linalg.eig(A)  
print(vals)  
print(V)  
print(A @ V[:,0] - vals[0]*V[:,0]) # near zero
```

Expected: eigenvalues 3 and 1; residual vector near (0,0).

Exercises

Pick Octave or Python for the term.

E1. Lab 4 mystery explained. Let

$$B = \begin{bmatrix} 1.2 & 0 \\ 0 & 0.5 \end{bmatrix}.$$

- (a) Use software to find eigenpairs of B .
- (b) Verify $Bv = \lambda v$ for each pair (hand or code).
- (c) In one sentence: why does $x_0 = (1, 0)$ grow while $y_0 = (0, 1)$ decays?

Expected: $\lambda = 1.2$ with $v \sim (1, 0)$; $\lambda = 0.5$ with $v \sim (0, 1)$. Growth vs decay follows from $|\lambda| > 1$ vs $|\lambda| < 1$ along those persistent directions.

E2. Verify a nondiagonal matrix. Let

$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

(from Lab 4 E1).

- (a) Compute eigenpairs with software.
- (b) For each pair, print $\|Av - \lambda v\|$ (or the residual vector). Accept “near zero.”
- (c) Which eigenvalue has larger absolute value? Predict the long-run direction of $A^t x_0$ for a generic x_0 .

Expected: eigenvalues 1 and 0.7 (exactly, up to rounding); dominant $\lambda = 1$ with eigenvector along (2, 1) (or parallel). Stabilization toward that mix matches Lab 4.

E3. Simulate and explain. Same A as E2, $x_0 = (1000, 0)$. Run about 20 steps (reuse Lab 4 code). Using E2’s eigenpairs, explain in 2–3 sentences why the state approaches a steady mix rather than blowing up or oscillating.

Expected: both $|\lambda| \leq 1$; the $\lambda = 1$ mode persists and the $|\lambda| = 0.7$ mode decays, so x_t settles along the $\lambda = 1$ eigenvector.

E4. Complex / oscillation. Let

$$R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

- (a) Software eigenvalues (expect near $\pm i$).

- (b) Compute $x_0 = (1, 0)$, $x_1 = Rx_0$, x_2 , x_3 , x_4 by hand or code.
- (c) In one sentence: how do complex eigenvalues relate to the cycling you see?

Expected: $\lambda = \pm i$; states cycle every 4 steps on the unit circle; complex eigenvalues encode pure rotation / oscillation (no real persistent direction).

E5. Symmetric / normal-mode peek. Let

$$S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

(symmetric; same as the demo).

- (a) Get both eigenvectors v_1, v_2 from software.
- (b) Compute $v_1^\top v_2$. Are they orthogonal (up to rounding)?
- (c) Normal modes (application/demo): for a symmetric vibration-style matrix, orthogonal eigenvectors are the independent “shapes” that stay aligned under S . No differential-equation derivation required — just observe the orthogonality.

Expected: $v_1^\top v_2 \approx 0$; yes, orthogonal.

E6 (stretch). Dominant growth. Let $C = \begin{bmatrix} 1.1 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$ and $x_0 = (1, 1)$. Find eigenpairs; identify the dominant $|\lambda|$; simulate to $t = 15$ and comment whether $\|x_t\|$ grows roughly like $|\lambda_{\text{dom}}|^t$.

One-pager

Preview the [Lab 5 ticket](#); write on the hard copy in class.